

Dai volti ai gruppi: una mappatura multidimensionale della rappresentazione percettiva dei gruppi sociali nell'infanzia

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"In 2023, Member States issued more than 3.7 million residence permits to non-EU citizens from more than 170 countries."

https://www.europarl.europa.eu/doceo/document/E-10-2025-000294-ASW_EN.html

Social Categorization

Social categorization changes how people perceive, evaluate, remember, and treat one another (Amodio & Cikara, 2021; Levy et al., 2023; Rhodes & Baron, 2019)

The Outgroup Homogeneity Effect

Early Emergence of Ingroup favoritism vs. Childhood OHE

- **Ingroup favoritism:** Strong evidence for ingroup preferences and visual specialization for ingroup faces (Kelly et al., 2009; Quinn et al., 2016).
- **OHE in childhood:** No consolidated evidence for the Outgroup Homogeneity Effect (OHE). Why?

A few structural obstacles

1. Difficult not to employ explicit tasks (creating top-down categories)
2. Most of the literature employs a dyadic paradigm (ingroup-outgroup) due to the difficulty of creating multigroup contexts
3. If visual paradigms are used, the stimuli selection is subject to biases

1. Spontaneous bottom-up categorization

Multidimensional Scaling (MDS)

- Face-Space Theory (Valentine et al., 2016): Faces exist in a continuous multidimensional perceptual space
- **Goal:** Visually map whether faces from different ethnic groups are spontaneously perceived as more similar to one another (OHE), and how this changes across age groups.

2. Multi-group context

- 4-group context: Italian majority-ingroup + 3 distinct ethnic minority-outgroups (African Descent, East Asian, Andean)
- 20 faces per gender group (5 per ethnic group)

3. The AI generated-stimuli

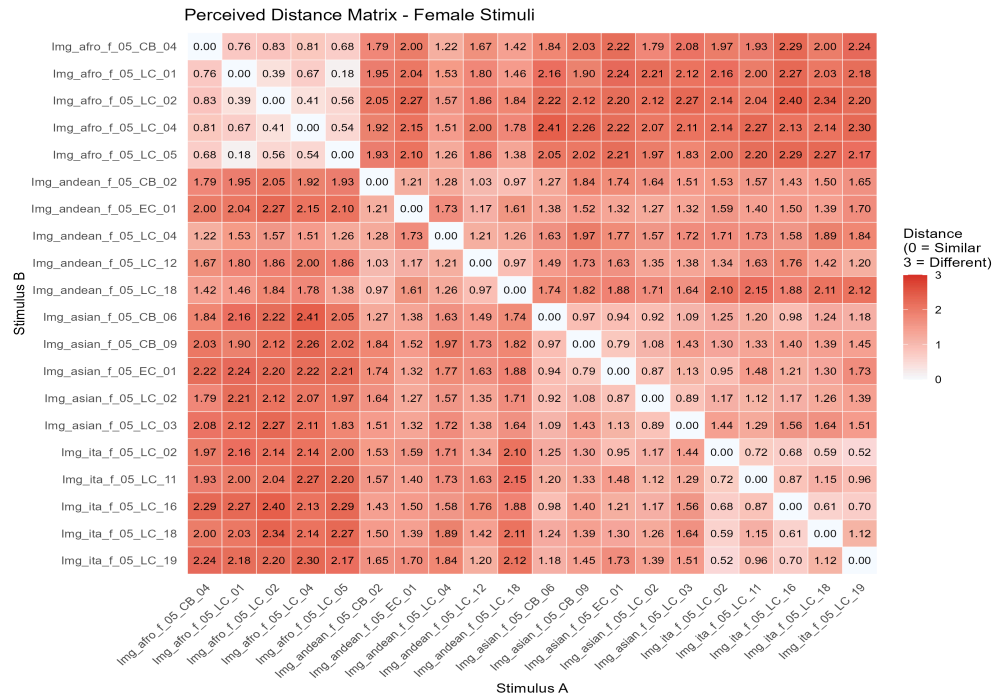
- Morphologies controlled using an emotional filter and FaceNet embeddings to ensure similar perceptual distinctiveness across groups

African Descent	East-Asian	Italian	Andean
			

→	→	→
+		Quanto si somigliano? Molto Diversi Un po' Diversi Un po' Simili Molto Simili
800ms	1000ms	//

The Face Similarity Task

- Incomplete block design
- 66 random dyads presented sequentially (3 faces per group)



The Trait Attribution Task

- 4 positive (Buoni, Belli, Simpatici, Puliti) and 6 negative (Cattivi, Brutti, Antipatici, Sporchi) traits
- 5 possible triads



Pre-registered Study

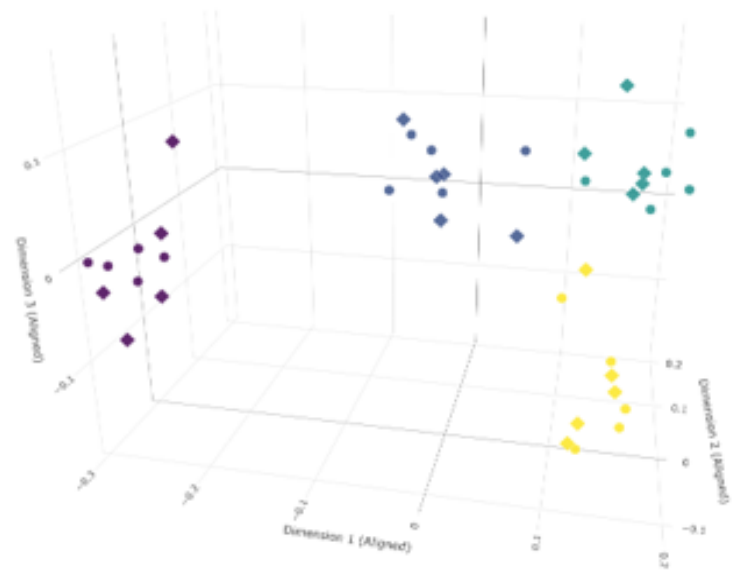
Sample:

- N=249, M=5.17 years, SD=1.23, 134 boys (53.8%) and 115 girls (46.2%)
- Age range: 3 to 8 years

MDS

3D MDS visualization

Cross-Study 3D Space
Male Targets: S1 (Circles) vs S2 (Diamonds)



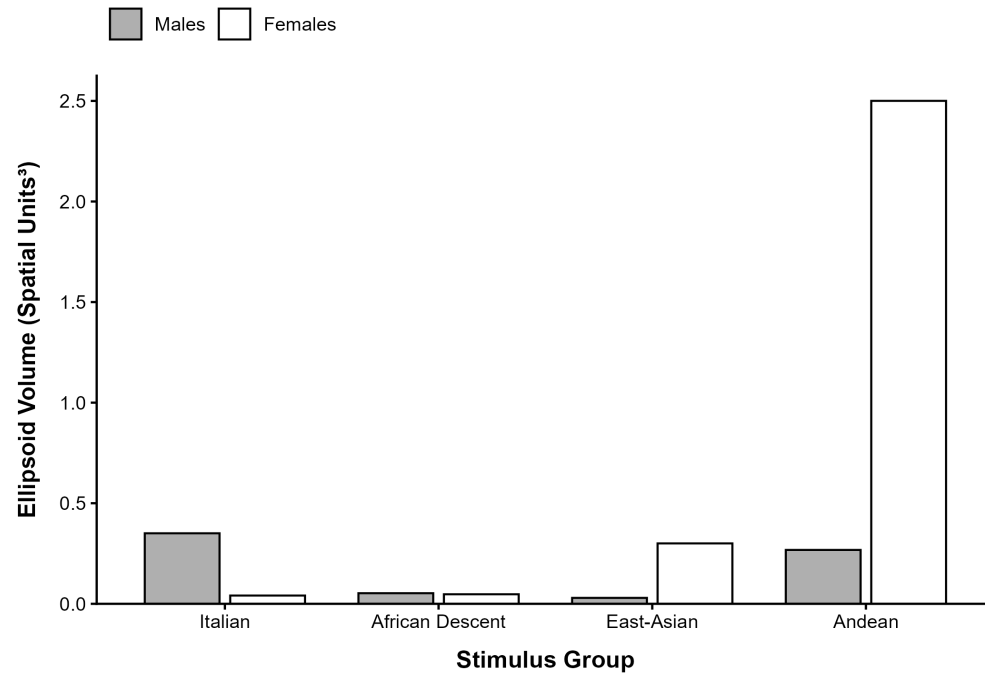
Legend

- African
- Study 1 (Moderated)
- African
- Study 2 (Unmoderated)
- Andean
- Study 1 (Moderated)
- Andean
- Study 2 (Unmoderated)
- East-Asian
- Study 1 (Moderated)
- East-Asian
- Study 2 (Unmoderated)
- Italian
- Study 1 (Moderated)
- Italian
- Study 2 (Unmoderated)

- Children spontaneously visually categorized stimuli into ethnic groups

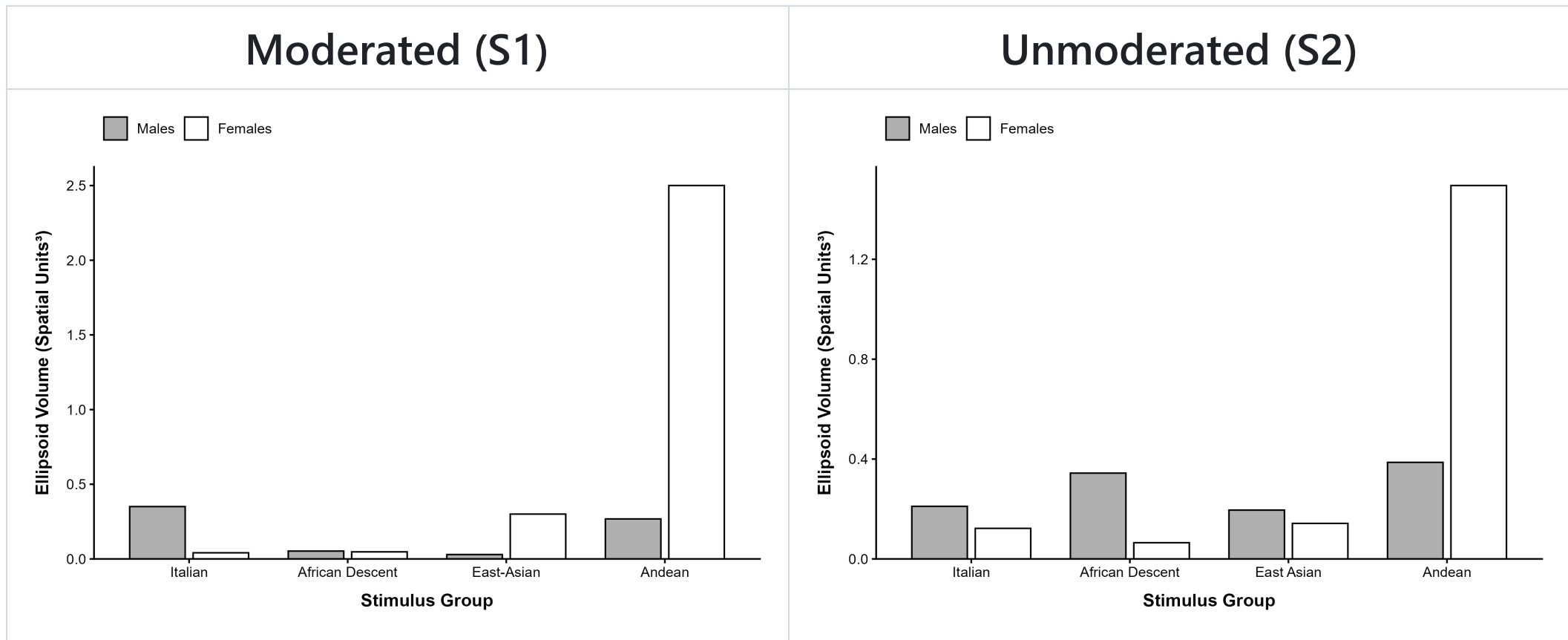
Outgroup Homogeneity Effect

- Ellipsoid's volume:



Outgroup Homogeneity Effect

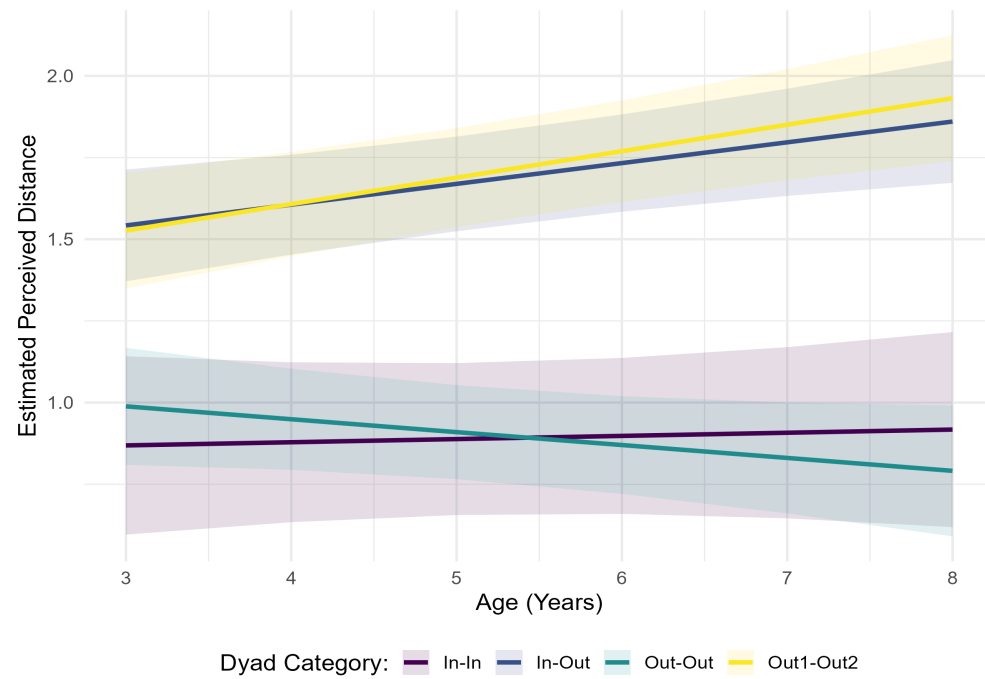
- Ellipsoid's volume:



- **No support for OHE**

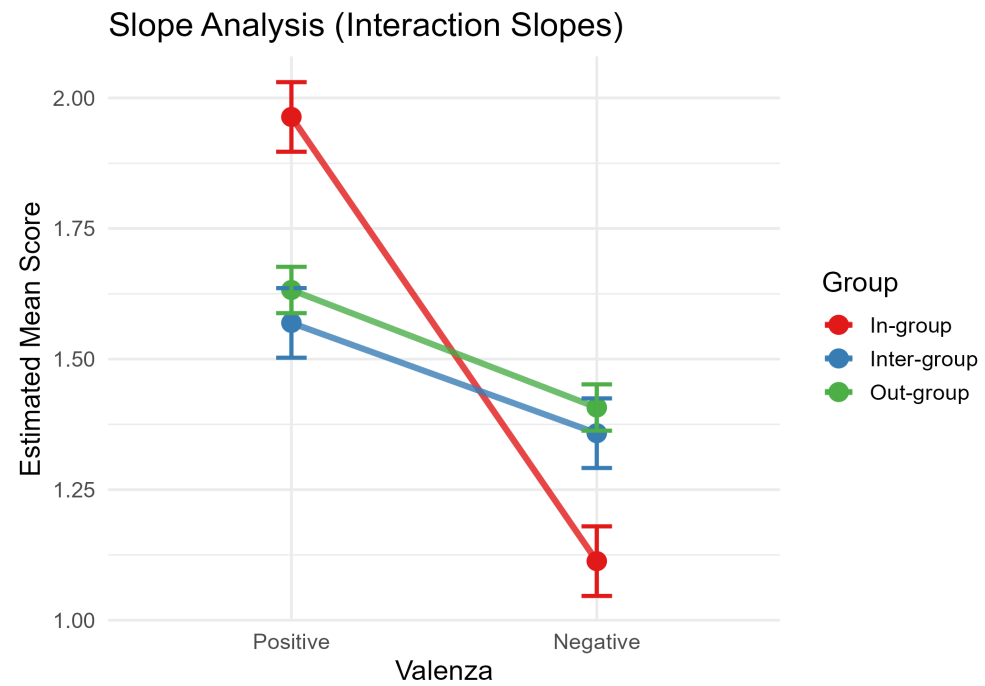
LMM - Developmental Trajectories

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- Intergroup dissimilarity judgments increased with age

Ingroup favoritism



- **Support for ingroup favoritism**

Conclusions

- Children spontaneously visually categorize stimuli into ethnic groups
- No support for OHE
- Intergroup dissimilarity judgments might increase with age
- Ingroup favoritism supported

Future directions

- Individual differences: Exploring how child-level factors (e.g., social contact, intergroup attitudes) shape perceptual dispersion.
- Task & contextual effects: Investigating whether the number and composition of groups alter face-space geometry.
- Longitudinal trajectories: Tracking individual shifts in categorization and outgroup perceptual boundaries over time.

Questions? 🙌



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